IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received May 15, 2020, accepted June 18, 2020, date of publication July 8, 2020, date of current version July 29, 2020.

Digital Object Identifier 10.1109/ACCESS.2020.3007871

Reinforcement Learning Based Load Balancing for Hybrid LiFi WiFi Networks

RIZWANA AHMAD[1], (Student Member, IEEE), MOHAMMAD DEHGHANI SOLTANI[2], MAJID SAFARI[2], (Member, IEEE), ANAND SRIVASTAVA[1], (Member, IEEE), AND ABIR DAS[3]

[1]Department of Electronics and Communication, IIIT-D, New Delhi 110020, India
[2]Institute for Digital Communications, School of Engineering, The University of Edinburgh, Edinburgh EH8 9YL, U.K.
[3]Computer Science and Engineering Department, IIT Kharagpur, Kharagpur 721302, India

Corresponding author: Rizwana Ahmad (rizwanaa@iiitd.ac.in)

ABSTRACT Light fidelity (LiFi) is an emerging communication technology, which utilizes the light-emitting diodes (LEDs) for high-speed wireless communications. Due to its huge unlicensed bandwidth, LiFi is capable of supporting high data rates. The quality of the LiFi channel fluctuates across the room due to interference, reflection from walls or blockage. On the other hand, WiFi is another wireless communication technology that is capable of providing moderate data rates with ubiquitous coverage. As the electromagnetic spectrum of LiFi does not overlap with WiFi, both of them can coexist to form a hybrid LiFi and WiFi network for seamless and high-throughput connectivity. The performance of a hybrid system significantly depends upon the access point (AP) assignment and resource allocation strategies. In this paper, a downlink hybrid system with one WiFi AP and four LiFi APs is considered, and a reinforcement learning (RL) algorithm is implemented in order to determine an optimal AP assignment strategy, which maximizes the long-term system throughput while ensuring the required users fairness and satisfaction. Furthermore, two different scenarios based on the random waypoint model with uniform and non-uniform distribution of users have been studied. The performance of the proposed system is compared against state-of-the-art benchmark approaches e.g., signal strength strategy (SSS), exhaustive search, and an iterative optimization method. The results are reported in terms of the average system throughput, user satisfaction, fairness, and capacity outage probability. It is shown that the proposed RL method performs closer to the exhaustive search scheme at fairly low complexity. The RL method also outperforms the SSS scheme and the iterative algorithm in most scenarios.

INDEX TERMS Hybrid LiFi WiFi, light fidelity (LiFi), load balancing, reinforcement learning (RL), trust region policy optimization (TRPO).

I. INTRODUCTION

There is an exponential increase in the wireless data requirement; cisco has reported that by 2022, the global mobile data traffic will reach 396 exabytes (396 billion GB) per month, that is the mobile data traffic will increase seven-fold from 2017 [1]. This rapidly growing data requirement has created a massive load on the current radio frequency (RF) based communication network. In order to meet future data requirement, many researchers in academia and industry have started investigating the visible light communication (VLC) for the indoor environment. VLC offers various advantages over RF, such as huge spectrum in license-free band, spatial reuse, and inherent security [2]–[4]. Furthermore, VLC neither causes health hazards nor interferes with other electromagnetic devices [5]. VLC is based on intensity modulation (IM) and direct detection (DD) technique, i.e., the information is transmitted by modulating the intensity of the light signal and the receiver detects the variation in the received intensity in order to decode the transmitted signal. VLC utilizes the light emitting diodes (LEDs) for transmission of the information signals and uses a photodetector (PD) at the receiver to convert the optical signal into an electrical signal. Light fidelity (LiFi) is the communication technology built over the physical layer of VLC, it is capable of supporting fully networked, bidirectional, and high-speed wireless communication, i.e., LiFi provides wireless networking extension of point-to-point VLC [6]–[8]. License-free deployment and

The associate editor coordinating the review of this manuscript and approving it for publication was Anandakumar Haldorai.

nearly universal availability of LEDs make LiFi an attractive and inexpensive choice for communication. Moreover, it has been shown that LiFi can achieve data rates in the range of Gbps [9], [10]. However, light cannot penetrate opaque objects due to which LiFi is prone to blockages. Therefore, the throughput achieved by LiFi fluctuates spatially leading to possible outage in many spatial areas of an indoor environment. It is important to note that LiFi can support high data rates when signal is sufficient, whereas WiFi can support moderate data rates with more ubiquitous coverage. Therefore, in order to accomplish a better quality of services (QoS), the hybrid integration of LiFi and WiFi is suggested [11]. Due to the non-overlapping spectrum of WiFi and LiFi, these two technologies can co-exist without interfering with each other [12]. Therefore, the resultant hybrid network provides higher system throughput as compared to standalone LiFi or WiFi networks [13]. It has been reported in the literature that an appropriately designed hybrid LiFi WiFi network can provide higher data rate, better user satisfaction, and better outage performance [14]–[16]. Efficient load balancing (LB) is a major challenge for hybrid LiFi WiFi networks [17]; the two main issues are user access point (AP) selection and resource allocation. For hybrid networks, AP selection is a complex process as the coverage area of LiFi and WiFi network overlaps with each other and WiFi covers larger area but has lower capacity, thus, WiFi is more prone to overload. When the traditional maximum received signal strategy (RSS) is used for AP selection in Hybrid LiFi WiFi network, WiFi AP becomes susceptible to overload, LiFi APs close to WiFi AP will be underused, and the system cannot ensure the required QoS [18]. Various studies have been conducted for efficient AP assignment in hybrid LiFi WiFi network which are summarised below.

A. RELATED WORK

The problem of AP assignment in hybrid LiFi WiFi network has been addressed by different methods, which can broadly be classified into three categorise, namely, optimization, fuzzy logic (FL) and machine learning. Different studies have used optimization based method in order to achieve different objectives in hybrid LiFi WiFi notworks. In [18], user mobility and handover signalling overheads are taken into account, and the utility function considered both the system throughput and fairness. Further, in this work, two different cases, namely, with and without optical co-channel interference were considered. In [19], the optimization problem for maximizing the achievable user QoS with certain outage probability is formulated and solved by two different optimization algorithms that jointly and separately optimize the resource allocation and AP assignment. In [20], a novel Load balancing method is proposed for optimizing the network throughput over a period of time instead of maximizing the instantaneous network throughput. In [21], authors have formulated a power allocation optimization problem for the hybrid LiFi WiFi scenario, under common backhaul constraint. In [22], the authors formulated an optimization problem in terms of power and bandwidth allocation in order to maximize the energy efficiency of a heterogeneous VLC RF communication system. In [23], the authors have presented unified resource allocation and mobility management algorithm for indoor VLC networks based upon particle swarm optimization. It is important to note that all these schemes are formulated as optimization problems to be solved at every time-step, therefore, resulting into a high computational complexity.

Other studies have used FL method to assign APs to users. In [24], authors have explored FL based dynamic load balancing scheme in order to mitigate the handover effects. This scheme considers the desired data rate and speed of user, to determine whether a handover is required or not. The user speed information is exploited by FL, so that more suitable APs can be assigned to users moving at a fast speed or users experiencing transient shadowing effect. Furthermore, by using FL ping-pong pattern of handover was avoided. In [25], two-stage AP selection method was proposed, the first stage determines the association of users to WiFi AP based on FL, and the second stage assigns the remaining users to the LiFi network. The results achieved by this method was close to the optimization based method in terms of throughput, at a significantly reduced complexity.

Some studies have explored machine learning based approaches to solve the problem of AP assignment in hybrid LiFi WiFi Network. In [26], authors proposed the concept of responsive association (RA) where the associations are established by taking into account both the users' current geo-locations and their current queue backlog states. Furthermore, they considered the concept of anticipatory association (AA), where the associations are established by taking into account both the users' time-variant geo-locations and their evolving queue backlog states. They have reported the results in terms of delay versus throughput trade-offs. In [27], authors proposed transfer learning based network selection algorithm, i.e., reinforcement learning with knowledge transfer. Specifically, in this work, context information is leveraged to tackle the network selection on two aspects. On one hand, the asymmetric uplink and downlink performance requirements of traffic are explicitly modeled. On the other hand, some distinguishing features of network as well as the stationary distribution law of network load are used to assist the algorithm design. This work is different from proposed work as reinforcement learning algorithm used is Q-Learning with knowledge transfer whereas in our work we have used trust region policy optimization (TRPO) method. Moreover, this work only reported the improvement in the reward which is a function of instantaneous uplink and downlink throughput. However, the QoS parameters: user's satisfaction and fairness has not been studied in this paper. In [28], authors presented AP selection strategies for hybrid LiFi WiFi network based on the multi-armed bandit, where the decision probability distribution is updated based on the 'exponential weights for exploration and exploitation' algorithm and the 'exponentially weighted algorithm with linear programming' algorithm. The proposed work is different from the earlier work as the it uses single agent gradient decent based learning as compared to multi-agent being used in the previous work.

B. MAIN CONTRIBUTIONS

Motivated by these earlier works, in this paper, we have used reinforcement learning (RL) based method for performing the load balancing in a hybrid LiFi WiFi network. It has been shown that RL based load balancing provides improved average network throughput, user satisfaction, fairness and outage performance. The system is analyzed under two different user behaviour.

The main contributions of this paper are summarized as follows:

- We have proposed RL based dynamic load balancing for hybrid LiFi WiFi network. The reward function of RL has been designed not only to maximize the long-term average network throughput but also to improve both users satisfaction and fairness.
- The results are compared against the state-of-the-art signal strength strategy (SSS), an iterative optimization method and exhaustive search, explained section in III-B. The results are presented in terms of computational complexity, average network throughput, user satisfaction, fairness and capacity outage probability.
- Two different scenarios with different user behavior models have been considered to demonstrate the adaptability of the proposed RL method, they are referred as random waypoint (RWP) and hotspot random waypoint (HRWP), explained in section II-C.

The rest of the paper is organized as follows. The system model is described in Section II and Section III introduces the proposed RL based load balancing methods. The performance evaluation and discussion are presented in Section IV, and the paper is concluded in Section V.

II. SYSTEM MODEL

In this paper, a multi-user hybrid LiFi WiFi network is considered for indoor communication with four LiFi APs and one WiFi AP, as shown in Fig. 1. In our work, we have represented the total number of APs and users in system by N_{AP} and N_{u}, respectively. A typical room of $5 \times 5 \times 3$ m^3 is considered with the WiFi AP is located at centre of the room to provide coverage to the entire room. The coverage of each LiFi AP is confined to a smaller area, known as attocell. In this system, a central unit (CU) is connected to both WiFi and LiFi APs which is responsible for making the load balancing decisions. It is assumed that CU has access to necessary information through error-free feedback link. As all LiFi APs reuse the same modulation bandwidth, therefore, users in overlapping attocell areas may experience optical interference, which is treated as additional noise in this paper. The spectrum reuse in LiFi provides advantage of higher spatial spectral efficiency at the cost of significantly low data rate in the overlapping attocell areas. In addition, the users are uniformly distributed across the room and they follow random way point and hotspot random waypoint models explained in II-C for mobility. Furthermore, we have also assumed that the users are accessing the internet for HD videos, hence they require higher data rates which is modeled as a Poisson process with parameter set to 50 Mbps. Moreover, in this work, each user can connect to a single AP at a time (i.e., either a LiFi AP or the WiFi AP). For the sake of simplicity, we have considered time-division multiple access (TDMA) to support multiple users connection to single AP, and round robin based method for resource allocation in this work.

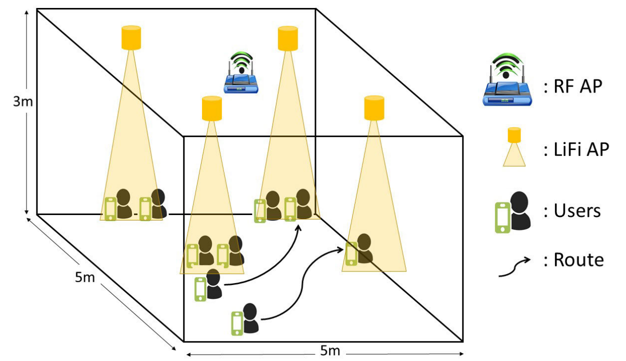


FIGURE 1. Schematic diagram of a hybrid LiFi WiFi network.

A. LiFi CHANNEL MODEL

The LiFi optical channel consists of two components, namely, line-of-sight (LOS) and non line-of-sight (NLOS). The LOS channel gain [25] is given by:

$$H_{\mathrm{LOS}} = \frac{(m+1)A_{\mathrm{PD}}}{2\pi d^2} \cos(\phi) g_{\mathrm{f}} g_{\mathrm{c}}(\psi) \cos(\psi) \tag{1}$$

where A_{PD} is the physical area of PD, m is the Lambertian order of the transmitter, d is the distance between LiFi AP and user, g_{f} denotes the gain of the optical filter; and g_{c} defines the gain of optical concentrator, and is given as:

$$g_{\mathrm{c}}(\psi) = \begin{cases} \dfrac{n^2}{\sin^2(\psi)}, & 0 \leq \psi \leq \Psi \\ 0, & \psi > \Psi \end{cases} \tag{2}$$

where n is the refractive index and Ψ is the semi-angle of field of view (FOV) of PD. The channel gain for NLOS component is defined as [29]:

$$H_{\mathrm{NLOS}} = \frac{\rho A_{\mathrm{PD}} e^{j2\pi f \Delta T}}{A_{\mathrm{room}}(1-\rho)(1+j\frac{f}{f_c})} \tag{3}$$

where ρ is the reflectivity of the walls, ΔT is the delay between the LOS and diffused signals, A_{room} is the area of the room, and f_c is the cut-off frequency. In this work, we have considered the constant reflectivity for the reflecting services, but the same channel model is capable of incorporating the non-constant reflectivity from different surfaces, as well. However, this modelling does not change the conclusion of

TABLE 1. LiFi channel parameters.

Channel Parameter	Symbol	Value
Height difference between the user and AP	h	2 m
Area of PD	A_{PD}	1 cm^2
The gain of opitcal filter	g_{f}	1
Half intensity radiation angle	$\theta_{1/2}$	60°
Field of View (FOV) of PD	Ψ	60°
Responsivity	$\mathcal{R}_{\mathrm{PD}}$	0.53 A/W
Reflection coefficient	ρ	0.8
Transmit optical power per LiFi AP	P_{opt}	3 Watt
Bandwidth per LiFi AP	B_{LiFi}	40 MHz
PSD of LiFi noise	N_{LiFi}	10^{-21} A^2/Hz

this study. Therefore, for the sake simplicity, we have considered constant reflectivity model in our work. The complete optical channel is given as $H_{\mathrm{LiFi}} = H_{\mathrm{LOS}} + H_{\mathrm{NLOS}}$. The LiFi channel parameters used in simulation are stated in Table 1 [24], [25]. The signal-to-noise ratio (SNR) for the user μ connected to LiFi AP α is represented as $SNR_{\mu,\alpha}$, and can be expressed as:

$$SNR_{\mu,\alpha} = \frac{(H_{\mathrm{LiFi}(\mu,\alpha)} P_{\mathrm{opt}} \mathcal{R})^2}{N_{\mathrm{LiFi}} B_{\mathrm{LiFi}}}, \tag{4}$$

where $\mathcal{R}$ represents responsivity of PD, $H_{\mathrm{LiFi}(\mu,\alpha)}$ is the channel gain between AP α and user μ, P_{opt} denotes transmitted optical power, N_{LiFi} is the LiFi noise power spectral density (PSD), B_{LiFi} is the bandwidth of LiFi AP. It is important to note that as the same frequency is reused in each LiFi AP, therefore, the signal received from other than intended LiFi AP is perceived as interference. Thus, the signal-to-interference-noise ratio (SINR) for the user μ connected to LiFi AP α is denoted by $SINR_{\mu,\alpha}$ and can be expressed as:

$$SINR_{\mu,\alpha} = \frac{(H_{\mathrm{LiFi}(\mu,\alpha)} P_{\mathrm{opt}} \mathcal{R})^2}{N_{\mathrm{LiFi}} B_{\mathrm{LiFi}} + \sum_{\beta \in AP} (H_{\mathrm{LiFi}(\mu,\beta)} P_{\mathrm{opt}} \mathcal{R} \kappa)^2} \tag{5}$$

where $H_{\mathrm{LiFi}(\mu,\beta)}$ is the channel gain between interfering LiFi APs β and the user μ. The achievable data rate between LiFi AP α and user μ, is calculated by the capacity lowerbound [30], [31] and is given as:

$$r_{\mu,\alpha} = \frac{B}{2} \log_2 \left(1 + \left(\frac{6}{\pi e}\right) SINR_{\mu,\alpha}\right). \tag{6}$$

B. WiFi CHANNEL MODEL

The WiFi channel gain is given by [19]:

$$G_{\mu,\alpha}(f) = \sqrt{10^{\frac{-L(d)}{10}}} h_{\mathrm{r}} \tag{7}$$

where h_{r} and $L(d)$ represents the small-scale fading gain and the large-scale fading loss (in dB), respectively and f is the carrier frequency. The small-scale fading gain h_{r}, follows independent identical Rayleigh distribution with an average power of 2.46 dB and $L(d)$ [19] is given by:

TABLE 2. WiFi channel parameters.

Channel Parameter	Symbol	Value
Breakpoint distance	d_{BP}	5 cm
Shadowing loss	X_{SF}	3 dB
Central carrier frequency	f_c	2.4 GHz
Transmit Power	P_{WiFi}	20 dBm
Bandwidth per WiFi AP	B_{WiFi}	20 MHz
PSD of noise	N_{WiFi}	-174 dBm/Hz

$$L(d) = \begin{cases} L_{\mathrm{FS}}(d) + X_{\mathrm{SF}}, & d < d_{\mathrm{BP}} \\ L_{\mathrm{FS}}(d_{\mathrm{BP}}) + 35\log(\frac{d}{d_{\mathrm{BP}}}) + X_{\mathrm{SF}}, & d \geq d_{\mathrm{BP}} \end{cases} \tag{8}$$

where, d is the separation distance, L_{FS} is the free space loss, d_{BP} is breakpoint distance and X_{SF} is the shadowing loss. L_{FS} is given as $L_{\mathrm{FS}}(d) = 20\log_{10}(d) + 20\log_{10}(f) - 147.5$ (dB). The WiFi channel parameters used in simulation are stated in Table 2 [24], [25]. The SNR for the μ user connected to WiFi AP can be expressed as:

$$SNR_{\mu,\alpha}(f) = \frac{|G_{\mu,\alpha}|^2(f) P_{\mathrm{T}}}{N_{\mathrm{WiFi}} B_{\mathrm{WiFi}}}, \tag{9}$$

where P_{T} is the transmitted power, N_{WiFi} is PSD of noise in WiFi link, B_{WiFi} is the bandwidth of WiFi AP and gain $G(\mu,\alpha)(f)$ can be calculated using (7). Since only one WiFi AP is considered in this system, hence there will be no interference, and the SNIR for WiFi AP will be same as SNR, as given in (9). The achievable data rate between WiFi AP α and user μ, is calculated using Shannon's capacity and is expressed as:

$$r_{\mu,\alpha} = B \log_2(1 + SNR_{\mu,\alpha}). \tag{10}$$

C. RANDOM WAYPOINT MOBILITY MODEL

In general, most of the studies consider random way point (RWP) model for mobility. It is a simple model in which a user selects a random destination and moves towards that destination at a constant speed. After reaching the destination, user picks up a new destination and starts moving in that direction and this process continues. However, it is also interesting to consider that inside a room, there are hotspots, where the users would gather with high probability. For example, users may gather under the LEDs to have better illumination level.

In [32], authors have introduced a modified random waypoint model, that allows to create attraction points by modifying the distribution of destination points. The distribution of the destination points (x_d, y_d) is given by:

$$\begin{aligned} f(x_d, y_d) = &\frac{1}{A_{\mathrm{room}} + (\xi - 1)A_a} \times \Big[(u(x_d + x_m) \\ &- u(x_d - x_m))(u(y_d + y_m) - u(y_d - y_m)) \\ &+ (\xi - 1)(u(x_d - x_{a,max}) - u(x_d - x_{a,min})) \\ &\times (u(y_d - y_{a,max}) - u(y_d - y_{a,min}))\Big] \end{aligned}$$

where A_a is the area of attraction point, defined by $x_{a,min} \leq x \leq x_{a,max}$ and $y_{a,min} \leq y \leq y_{a,max}$, and ξ is the intensity.

In our work, we have considered four hotspots located around four LEDs in the room with $\xi = 200$ and $A_a = 0.25\text{m}^2$. Additionally, a wait time has been introduced i.e., once a user reaches to a hotspot, it will wait for some time at that particular location. Afterwards, the user will select a new destination and the process will continue. This particular mobility model will be referred as for hotspot random waypoint model (HRWP) in this paper. Fig. 2, shows the distribution of 100 nodes inside the room with above mentioned specifications for HRWP model.

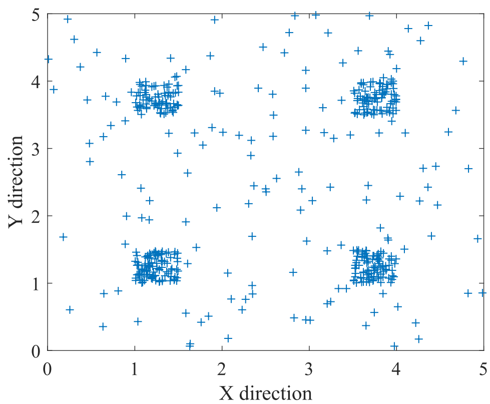


FIGURE 2. Distribution pattern of 100 nodes inside room for modified random waypoint model.

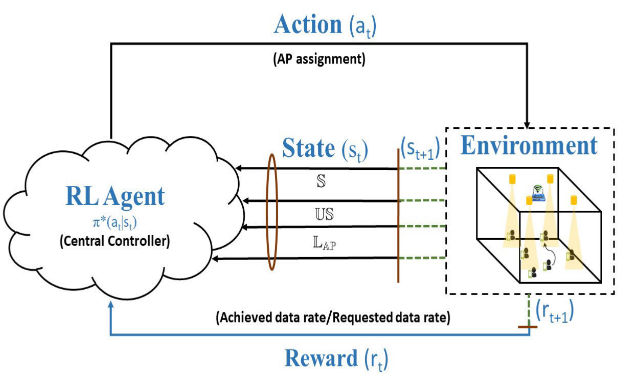


FIGURE 3. RL for a hybrid LiFi WiFi network.

III. PROPOSED LOAD BALANCING METHODS

A. REINFORCEMENT-LEARNING (RL) METHOD

RL is a promising machine learning approach, where, an agent continuously interacts with the environment in order to observe the state of the environment and to take actions based on these observations, as shown in Fig. 3. The ultimate goal of RL agent is to determine a stochastic policy, which maps states to a probability distribution over actions, in order to maximize the cumulative reward. RL can be viewed as an stochastically optimized solution for Markov Decision Processes (MDPs), when the MDP is not known. In the standard formulation of MDP, at time step $t \geq 0$, an agent is in state $s_t \in S$, takes an action $a_t \in A$, receives an instant reward $r_t = r(s_t, a_t) \in R$ and transits to a next state $s_{t+1} \sim P(.|s_t, a_t) \in S$. $\pi : S \rightarrow P(A)$ is the policy, where $P(A)$ is the set of distributions over the action space A [33]. The discounted cumulative reward under policy π is

$$\eta(\pi) = \mathbb{E}_{s_0,a_0,\ldots}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)\right],$$
$$s_0 \sim \rho_0(s_0), \quad a_t \sim \pi(a_t|s_t), \; s_{t+1} \sim P(s_{t+1}|s_t, a_t)$$

where $\gamma \in (0, 1)$ is the discount factor where lower values of γ means more emphasis on immediate rewards. The goal of RL is to find an optimal policy, π^*, which achieves the maximum $\eta(\pi)$ i.e.,

$$\pi^* = \underset{\pi}{\text{argmax}}(\eta(\pi))$$

We have formulated the given hybrid LiFi WiFi network AP assignment problem as an infinite-horizon discounted MDP defined by $(S, A, P, r, \rho_0, \gamma)$ where

- S is the set of continuous states, which includes SNR for all the users from all the APs represented by a matrix $\mathbb{S}$, the current load on each AP ($\mathbb{L}_{\mathbb{AP}}$) i.e., number of users connected to a particular AP out of total number of users, and user satisfaction represented as $\mathbb{US}$. The dimensions of $\mathbb{S}$ is $[N_\text{u} \times N_\text{AP}]$, dimensions of ($\mathbb{L}_{\mathbb{AP}}$) is $[N_\text{AP}]$ and dimensions of $\mathbb{US}$ is $[N_\text{u}]$. Therefore, dimension of the observation space is $[N_\text{u} + N_\text{AP} + N_\text{u} \times N_\text{AP}]$.
- A is a finite set of discrete actions, which is AP assignment, therefore, the dimension for discrete action space is $[N_\text{u}]$.
- $r : S \times A \rightarrow \mathbb{R}$ is the reward function, which is $(\frac{r_{\mu,\alpha} k_{\mu,\alpha}}{R_\mu})$, where $r_{\mu,\alpha}$ is the achievable data-rate between AP α and user μ, which is given by Eq. 6 and Eq. 10, R_μ is the required data rate of user μ and $k_{\mu,\alpha}$ represents the time slot allocation between AP α and user μ which is given as:
$$k_{\mu,\alpha} = \frac{1}{\sum_\alpha g_{\mu,\alpha}}, \tag{11}$$
where $g_{\mu,\alpha}$ is a binary variable which indicates the connection between AP α and user μ, i.e., $g_{\mu,\alpha} = 1$ indicates user μ is connected to AP α, otherwise not. The reward function is carefully designed to take into account both the throughput and user satisfaction.
- The discount factor γ is set to 0.9, $P : S \times A \times S \rightarrow \mathbb{R}$ is the transition probability distribution, $\rho_0 : S \rightarrow \mathbb{R}$ is the distribution of the initial state s_0.

In this work we have used a multi-layer perceptron (MLP) as policy network with parameters θ, for the task of access point assignment. Thus, the policy π can be written as $\pi(a_t|s_t; \theta)$. The state-action value function Q_π, value function V_π, and the advantage function A_π are defined as:

$$Q_\pi(s_t, a_t) = \mathbb{E}_{s_{t+1},a_{t+1},\ldots}\left[\sum_{l=0}^{\infty} \gamma^t r(s_{t+l}, a_{t+l})\right],$$
$$V_\pi(s_t) = \mathbb{E}_{a_t,s_{t+1},\ldots}\left[\sum_{l=0}^{\infty} \gamma^t r(s_{t+l}, a_{t+l})\right],$$
$$A_\pi(s, a) = Q_\pi(s, a) - V_\pi(s),$$

where $a_t \sim \pi(a_t|s_t; \theta)$, $s_{t+1} \sim P(s_{t+1}|s_t, a_t)$ for $t \geq 0$.

The training process involves the optimization of policy parameter θ of the MLP in order to maximize the expected discounted return. For this purpose, Trust Region Policy Optimization (TRPO) is used, which is a scalable algorithm for optimizing policies by using gradient descent [34]. These policy gradient methods are model free and provides better training stability than value iteration methods [35].

Moreover, TRPO provides guaranteed monotonic improvements under certain assumptions [34], which means the policy update in each TRPO iteration creates a better policy. TRPO methods directly learn the policy $\pi(a|s;\theta)$ using gradient descent, while constraining the update size of θ at each step. In particular, TRPO enforces this by introducing a KL divergence constraint on the size of update between the old and new policy at each iteration. The optimization problem in TRPO can be written as:

$$\begin{aligned} &\underset{\theta}{\text{maximize}}\ \mathbb{E}_{s\sim\rho_{\theta_{old}},a\sim q}\left[\frac{\pi_\theta(a|s)}{q(a|s)}Q_{\theta_{old}}(s,a)\right] \\ &\text{subject to } \mathbb{E}_{s\sim\rho_{\theta_{old}}}\left[D_{\mathrm{KL}}(\pi_{\theta_{old}}(.|s)||\pi_\theta(.|s))\right] \leq \delta \end{aligned} \tag{12}$$

where δ is a tunable parameter.

Algorithm 1 TRPO Algorithim to Find Optimal Policy

Input: Initial policy parameter, θ_0
Output: Optimal policy, $\pi^*_\theta(a_t|s_t)$

for k = 0, 1, 2, . . . **do**
  Run policy π_θ to collect the set of trajectories
  Estimate the advantage function $A_{\pi k}$
  Compute policy gradient g_k and KL-divergence Hessian-vector product function $f(v) = H_k v$
  Use conjugate gradient to calculate $x_k \approx {H_k}^{-1}{g_k}^{-1}$
  Estimate step update $\Delta_k \approx \sqrt{\frac{2\delta}{x_k^T H_k x_k}}x_k$
  Update step θ_{k+1}, using line search.
  $\theta_{k+1} = \theta_k + \alpha^k\Delta_k$
end for

Algorithm 2 Algorithm for Load Balancing Using RL

Input: Current state of hybrid LiFi WiFi network, s_t and optimal policy $\pi^*_\theta(a_t|s_t)$
Output: Reward r_{t+1} and state s_{t+1}

Use $\pi^*_\theta(a_t|s_t)$ for given s_t, in order to predict a_t
Update the AP assignment in the environment based on a_t.
Based on new AP assignment calculate reward r_{t+1} and new state s_{t+1}.
return r_{t+1} and s_{t+1}

The exact steps required to determine the optimal policy $\pi^*_\theta(a_t|s_t)$ are explained in algorithm 1. At the end of the training process an optimal stochastic policy $\pi^*(a|s;\theta)$ is obtained, which provide a probability distribution between the state and action, i.e., given a particular state the policy returns the probability of choosing a particular action. This policy can be used in real time for predicting the appropriate AP assignment, calculating the reward and the next state, as explained in algorithm 2. The run time computational complexity of RL based system is usually smaller as compared to optimization based method. For fair comparison, we have defined the optimization problem with an objective function same as the reward function of the RL and discussed possible solutions in section III-B. In the next section, we will discuss about the training and convergence performance of the RL algorithm.

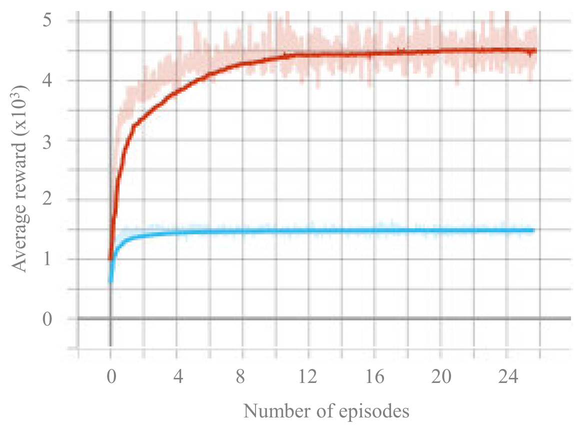


FIGURE 4. Training performance and convergence of RL in terms of average reward w.r.t number of episodes for $\gamma = 0.9$ (red) and $\gamma = 0.7$ (blue).

1) TRAINING PERFORMANCE AND CONVERGENCE OF RL

In this paper, we have used TRPO which imposes a trust region constraint on the policy to stabilize learning. For the sake of simplicity, we have reported the training results for 5 users, similar trend can be observed for 10 users. Fig. 4 shows the training performance of RL algorithm in terms of average reward for different discount factor, γ. It can be seen that the average reward of the algorithm converges after a certain number of episodes for both γ values. However, there is significant difference in value of average rewards obtained by the algorithm for different γ values. It can be observed that $\gamma = 0.9$ achieves average reward of around 3 times of average reward achieved by $\gamma = 0.7$. Therefore, $\gamma = 0.9$ has been chosen for simulation. However, the convergence of the algorithm for $\gamma = 0.9$ takes more time. In future work, we aim to include the concept of knowledge transfer in our algorithm to improve its convergence speed. TRPO being model-free algorithms, directly learns the optimal policy without estimating the model (transition and reward functions) of the environment. Therefore, TRPO requires relatively little tuning of hyperparameters; however, it suffers from higher sample complexity [36]. TRPO is an on-policy method, which means it explores by sampling actions according to the latest version of its stochastic policy. The exploration depends on both initial conditions and the training procedure. There are various methods proposed in literature to deal with the sample complexity of TRPO [37]–[39]; however, as this work is mainly focused on RL based load balancing for hybrid LiFi WiFi network, for the sake of simplicity, we have considered the simplest form of TRPO in this work. In future work, we aim to include the concept of knowledge transfer in our algorithm to improve its performance and convergence speed.

B. OTHER LOAD BALANCING METHODS

Inline with the objective of the proposed RL scheme, we can define an optimization problem for AP assignment in hybrid LiFi WiFi network as

$$\begin{aligned}
&\max_{g_{\mu,\alpha}k_{\mu,\alpha}} \sum_{\mu\in\mathbb{U}}\sum_{\alpha\in\mathbb{AP}} g_{\mu,\alpha}\log\left(\frac{r_{\mu,\alpha}k_{\mu,\alpha}}{R_\mu}\right)\\
&\text{s.t. } \sum_{\mu\in\mathbb{U}}(g_{\mu,\alpha}k_{\mu,\alpha}) = 1 \quad \forall\alpha\in\mathbb{AP}\\
&\qquad \sum_{\alpha\in AP} g_{\mu,\alpha} = 1 \quad \forall\mu\in\mathbb{U}\\
&\qquad g_{\mu,\alpha}\in\{0,1\}, \quad k_{\mu,\alpha}\in[0,1],\ \forall\mu\in\mathbb{U},\ \forall\alpha\in\mathbb{AP}
\end{aligned} \tag{13}$$

where $\mathbb{U}$ is the set of users, $\mathbb{AP}$ is the set of APs including both LiFi and WiFi APs, $r_{\mu,\alpha}$ is the achievable data-rate between AP α and user μ, which is given by (5) and (9), R_μ is the required data rate of user μ. The binary optimization variable $g_{\mu,\alpha}$ indicates the connection between AP α and user μ, i.e., if $g_{\mu,\alpha} = 1$, it means user μ is connected to AP α, otherwise not. The optimization variable $k_{\mu,\alpha}$ defines the time slot allocation between all the users connected to AP α. This optimization problem is a mixed integer non-linear programming (MINLP) problem which is mathematically intractable. Therefore, it is not possible to obtain a closed form solution for this problem.

There are three possible solutions:

- **Exhaustive search**: The exhaustive search provides a deterministic action at each step by systematically enumerating all possible actions for the given objective function and checking which particular action best satisfies the objective function. This ensures the best performance at the cost of high complexity. The exhaustive search implementation has been made possible for this problem because of the room dimension, which restricts the number of users and APs, therefore limits the computational complexity to a practical value. The objective function for exhaustive search is same as Eq. (13) and it has been considered in order to provide the upper bound performance at the cost of higher complexity.
- **Iterative algorithm**: The optimization problem given in Eq. (13) can be solved using an iterative algorithm [18]. By applying Lagrangian dual decomposition on the given objective function, the optimum value of $g_{\mu,\alpha}$ is calculated as:

$$g_{\mu,\alpha} = \begin{cases} 1 & \alpha = \operatorname{argmax}\left(\log(\frac{r_{\mu,\alpha}}{d_\mu}) - \lambda_\mu - \omega_\alpha\right)\\ 0 & \text{otherwise,}\end{cases} \tag{14}$$

  where λ_μ and ω_α are the Lagrangian's multipliers. By following the similar procedure as mentioned in [18], AP assignment and resource allocation is performed.
- **Signal Strength Strategy (SSS)** [25]: In a hybrid LiFi and WiFi network since the bandwidth and receiver noise differs for LiFi and WiFi, therefore, received signal strength does not fully represent the quality of channel. Hence, SNR is used as decision metric for SSS method. The objective function of the SSS method for a given user μ is defined as

$$\max_{\alpha}\ SNR_{\mu,\alpha} \quad \text{s.t } \alpha\in\mathbb{AP} \tag{15}$$

  where, $\mathbb{AP}$ is the set of APs including one WiFi and four LiFi APs and $SNR_{\mu,\alpha}$ represents the SNR values between μ user and α AP, which can be calculated by Eq. (9) and Eq. (4) for WiFi and LiFi APs, respectively.

IV. PERFORMANCE EVALUATION AND DISCUSSION

We have considered a typical 5 × 5 × 3 m indoor space which is entirely covered by a WiFi AP, and partially covered by 4 LiFi APs, as shown in Fig. 1. The current work is scalable to a larger room with more number of APs and users. However, as this work focuses on application of RL in order to obtain an optimal AP selection policy for hybrid LiFi WiFi network and we wanted to provide a performance upper bound using exhaustive search method, we have considered a commonly used scenario [20], [40]. It is assumed that all the optical attocells reuse the same bandwidth so there exists interference, which is treated as noise in the overlapping areas. Furthermore, we have also assumed that the users are accessing the internet for HD videos, hence they require higher data rates which is modeled as a Poisson process with parameter 50 Mbps. The users are assumed to be uniformly distributed across the room, and two different mobility models are considered, namely, RWP and HRWP, explained in section II-C. Moreover, for the sake of simplicity, we have assumed that each user can connect to a single AP at a time (i.e., either a LiFi AP or the WiFi AP) and TDMA to support multi-user connection. The RL agent is deployed on a CU which is connected to both WiFi and LiFi APs and is responsible for making the load balancing decisions. It is assumed that an error free feedback link exist to transmit user state information to CU. The simulation setup is coded in python 3.7 and an Open AI Gym environment has been built from scratch for the aforementioned hybrid LiFi WiFi network. We have used stable-baseline GitHub repository [41] for TRPO algorithm. The results reported are average over 200 episodes and the system parameters considered for simulation are summarized in Table 3.

The performance of the proposed RL method has been compared against exhaustive optimization, iterative optimization and SSS methods based on computational complexity, average network throughput, fairness, user satisfaction and capacity outage probability.

TABLE 3. System parameters.

System Parameter	Value
Room dimension	$5 \times 5 \times 3$ m^3
Number of APs	4 LiFi + 1 WiFi
WiFi AP location	(2.5 m, 2.5 m)
LiFi AP locations	($\pm$1.25 m,$\pm$1.25 m)
Number of Users, N_{u}	5,10
User distribution	Uniform
Mobility Model	RWP, HRWP
User Speed	1 m/s
Requested data rate, R_μ	Poisson with 50 Mbps
Multiple access technology	TDMA
RL Algorithm	TRPO
Gym environment	LiFi WiFi network
policy	MLP, 2 layers of 64
max KL divergence, δ	0.01
Discount factor, γ	0.9
Episode length, E	1000

A. PERFORMANCE METRICS

The three performance metrics namely, user satisfaction, fairness and capacity outage probability are defined as follows:

- The **users satisfaction** $S_{\mu,\alpha}$ is defined as the ratio of data rate achieved by the user to the data rate required by that user, it can be expressed as:

$$S_{\mu,\alpha} = \min\{1, \frac{k_{\mu,\alpha} r_{\mu,\alpha}}{R_\mu}\} \tag{16}$$

where R_μ is the requested data rate by user μ which is modeled as Poisson process. The maximum value of user satisfaction is 1, which implies that the user has achieved its requested data rate.

- For fairness, **Jain's fairness index** [42] is used. The user fairness index can be computed by [25]:

$$\eta = \frac{\left(\sum_{U} S_{\mu,\alpha}\right)^2}{N_{\text{u}} \sum_{U} (S_{\mu,\alpha})^2} \tag{17}$$

where N_{u} is the number of users.

- The **capacity outage probability** can be defined as:

$$\Phi = \Pr(k_{\mu,\alpha} r_{\mu,\alpha} < R_o) \tag{18}$$

where R_o denotes the threshold value for average required throughput.

B. COMPLEXITY ANALYSIS

In this work, we have used the Big-O notation for complexity as it provides a generalized upper bound or worst-case performance, which is independent of the underlying hardware. The SSS method simply selects the AP with highest SNR value out of total APs (N_{AP}). Therefore, its complexity is O($N_{\text{AP}} N_{\text{u}}$) [25]. The optimization method based on exhaustive search is computationally expensive as it searches over all

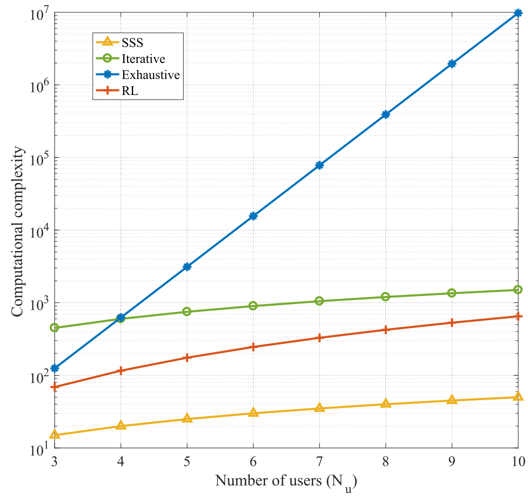


FIGURE 5. Computational complexity comparison of different schemes.

possible connections between users and APs, therefore, the complexity is O($(N_{\text{AP}})^{N_{\text{u}}}$). The complexity of optimization problem based on iterative methods depends on number of iterations I that it takes to converge. Its complexity can be estimated as O($N_{\text{u}} N_{\text{AP}} I$) [19]. The computational complexity of training phase of RL depends upon the number of states (n) in the MDP and it is given by O(n), however, this is just one time training cost. During the real time the complexity of RL algorithm depends on the optimal policy which is represented in form of MLP, the complexity depends upon the dimensions of observation and action space which in given in section III, therefore, the complexity for RL is O($N_{\text{u}}^2 N_{\text{AP}} + N_{\text{u}}^2 + N_{\text{u}} N_{\text{AP}}$). The complexity of SSS increases linearly with number of users, whereas for exhaustive optimization the complexity increases exponential with number of user. The complexity for iterative method is dependent on I and in case of RL the complexity has a quadratic relation with number of users. It is observed from Table 4 that the complexity of RL is much smaller as compared to exhaustive and iterative optimization. For better understanding, we have plotted the computational complexity of these methods for different number of users, assuming $I = 30$ [19], shown in Fig. 5. It can be observed from the graph that the complexity of RL is lower than exhaustive and iterative method.

TABLE 4. Computational complexity.

Scheme	Complexity
SSS	$O(N_{\text{AP}} N_{\text{u}})$
Iterative	$O(N_{\text{u}} N_{\text{AP}} I)$
Exhaustive	O($(N_{\text{AP}})^{N_{\text{u}}}$)
RL	O($N_{\text{u}}^2 N_{\text{AP}} + N_{\text{u}}^2 + N_{\text{u}} N_{\text{AP}}$)

C. EFFECT OF NUMBER OF USERS

The average network throughput for different number of users under various load balancing schemes are stated in Table 5. For 5 users, the iterative optimization based AP assignment provides 49.74 % increment in average network throughput

TABLE 5. Average network throughput (Mbps).

N_u	SSS	Iterative	Exhaustive	RL
5	116.23	174.04	239.90	218.81
10	73.12	98.72	138.41	110.69

over SSS, whereas the RL and exhaustive optimization provide an improvement of 88.26 % and 106.41 % over SSS in average network throughput respectively. As the number of users are increased to 10, RL and exhaustive search provides 51.37 % and 89.29 % improvement in average network throughput over SSS, respectively. The RL is able to provide significant improvement over SSS in throughput as compared to iterative algorithm for which the improvement in average network throughput over SSS is limited to 34.24 %. It is important to note that as the number of users increases, the network performance saturates, because of this the gain achieved by different load balancing methods is restricted.

TABLE 6. Fairness.

N_u	SSS	Iterative	Exhaustive	RL
5	0.9998	0.9999	1.0000	1.0000
10	0.9568	0.9743	0.9872	0.9727

The values of Jain's fairness index η for different number of users under various load balancing schemes are calculated using Eq. 17, they are tabulated in Table 6. For 5 users, all the schemes are able to achieve the fairness value very close to 1. However, for 10 users, SSS fairness is limited to 0.96, whereas both RL and optimization achieve a fairness of 0.97. The fairness index for exhaustive optimization was highest i.e., 0.97. Since the Jain's fairness index and average network throughput provide an average overview of the network performance. In order to provide a better insight on the performance of these schemes, the complementary cumulative distribution function (CCDF) of user satisfaction and capacity outage probability for particular throughput are provided.

Fig. 6 shows the CCDF of user satisfaction for different number of users under four different load balancing schemes. The value of satisfaction is calculated using Eq. 16. From the results, it can be observed that SSS based scheme performs worst in terms of user satisfaction, less than 40% and 25% of the users are completely satisfied for SSS based AP assignment method for 5 and 10 users, respectively. For iterative optimization, it is around 85% when there are 5 users but this value drops drastically to far less than 30% when the number of users are increased to 10. For exhaustive optimization and RL around 98% and 96% of users achieve complete satisfaction when there are 5 users, but as the number of users increases to 10, this value drops to around 45% and 40% respectively. Therefore, it can be observed that highest percentage of full user satisfaction is achieved by exhaustive search method followed by RL, which provides close enough performance. Moreover, an increase in number of users reduces the user satisfaction for all schemes.

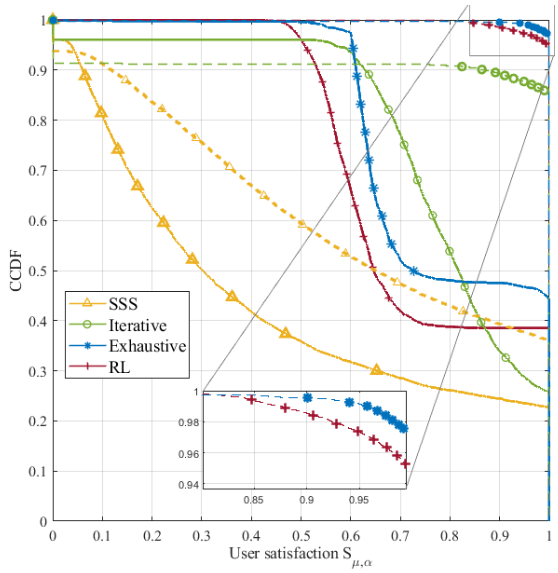


FIGURE 6. User satisfaction: N_u = 5 (dashed lines) and N_u = 10 (solid lines).

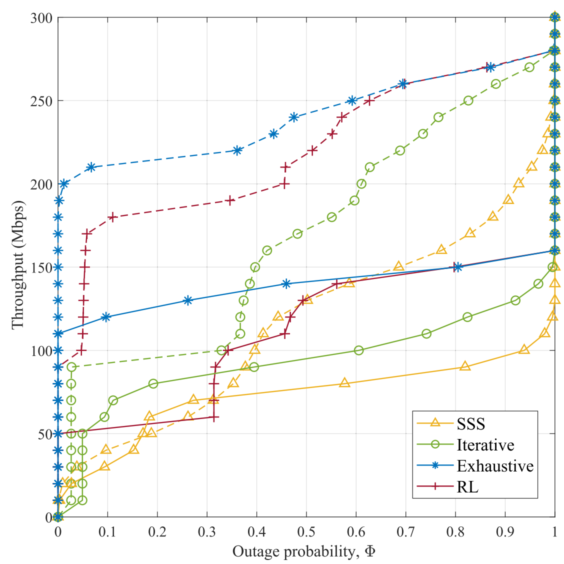


FIGURE 7. Capacity outage probability: N_u = 5 (dashed lines) and N_u = 10 (solid lines).

Fig. 7 illustrates the capacity outage probability for a threshold throughput of R_o. It can be observed that for 5 users, if a system allows 10% of the users to be in outage in terms of capacity, than the average network throughput can be around 210 and 180 Mbps for exhaustive search and RL respectively. In case of iterative algorithm this value is limited to less than 100 Mbps, where as for SSS it is around 40 Mbps. It can be observed that SSS performs worst, followed by iterative optimization. For larger capacity outage probabilities, beyond 0.6, the performance of RL and exhaustive optimization converges. Further, as the number of users increases from 5 to 10, for a particular capacity outage probability for the throughput value decreases significantly. Overall, exhaustive search provides highest possible throughput for a a

TABLE 7. Average network throughput (Mbps).

Mobility	SSS	Iterative	Exhaustive	RL
RWP	116.23	174.04	239.90	218.81
HRWP	122.23	197.28	267.86	214.85

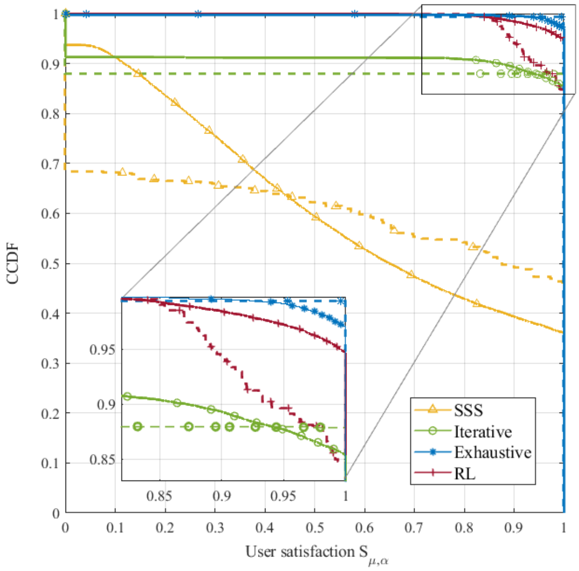


FIGURE 8. User Satisfaction: with RWP (solid lines) and HRWP (dashed lines).

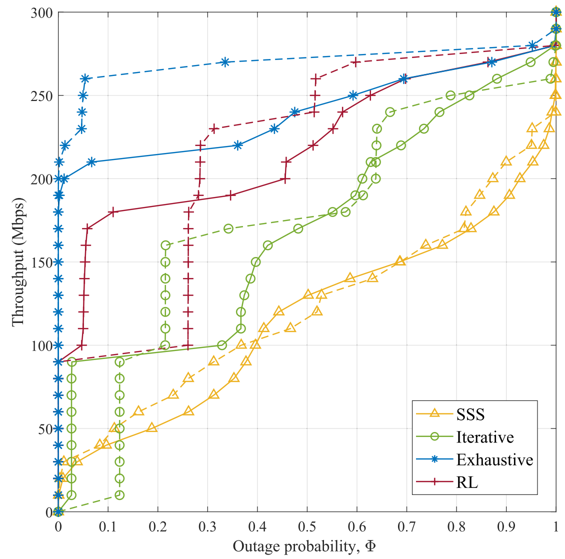


FIGURE 9. Capacity outage probability: with RWP (solid lines) and HRWP (dashed lines).

TABLE 8. Fairness.

Mobility	SSS	Iterative	Exhaustive	RL
RWP	0.9998	0.9999	1.00	1.00
HRWP	0.9990	0.9968	1.00	1.00

particular capacity outage probability, followed by RL, which provides almost similar performance at higher probability. Increase in the number of users degrades the performance for all schemes, but mostly RL performs close to exhaustive search.

D. EFFECT OF MOBILITY MODEL

In this work, we have also considered different user behaviours such as HRWP mobility model, which has four attraction points located around the four LEDs. Table 7 summaries the average network throughput for 5 users under two different mobility models. It is interesting to note that the pattern of performance remained the same for both mobility models. The exhaustive optimization provides the highest improvement in terms of average network throughput over SSS method, followed by RL and iterative optimization. The Jain's fairness index for 5 users under different mobility models are tabulated in Table 8. It can be observed that all the schemes are able to achieve full fairness ($\eta = 1$) for 5 users under both mobility models. Fig. 8 shows the CCDF of user satisfaction for 5 users under different mobility models. It can be observed that the user satisfaction changes significantly for the two mobility models. For SSS, in case of HRWP around 45% of the users are able to achieve full satisfaction whereas in case of simple RWP, it is only limited to 35%. Further, exhaustive search provides slight improvement in user satisfaction for HRWP. At the same time, RL provides performance matching to iterative optimization method for HRWP. Further, RL ensures 0.85 user satisfaction for all users for both mobility models, whereas iterative optimization scheme is only able to ensure the user satisfaction for 90% of the users, which means that at least 10% of users never achieves the desired user satisfaction for iterative algorithm. Moreover, SSS can ensure the user satisfaction only for 70% of the users in HRWP model. Fig. 9 illustrates the capacity outage probability of a particular throughput R_o for 5 users under two different mobility models. It can be observed that in case of HRWP the achievable throughput with certain capcity outage probability increases for exhaustive search. For SSS, the performance remained almost same with RWP and HRWP. Iterative algorithm and RL exhibits similar behaviour for the two mobility models, for smaller values of capacity outage probability, they support higher throughput in RWP models, but for larger values of outage probability, they support higher throughput in HRWP models. Overall,the trend of performance for all the four load balancing methods remains same for both the mobility models and it shows that RL provides robustness against different user behaviours.

V. CONCLUSION

In this paper, an RL based dynamic load balancing scheme has been considered for hybrid LiFi WiFi networks. The reward function of RL was carefully crafted to increase the long-term average system throughput while ensuring the required QoS. The RL was trained to determine the optimal policy, by using TRPO algorithm. Afterwards, the optimal policy was exploited to determine the appropriate AP assignment for the given state. Based on simulation results and complexity analysis, it is shown that the proposed method achieves a significantly better performance at lower run-time complexity compared to the conventional SSS scheme and the iterative algorithm. For 5 users and RWP mobility model, the RL algorithm is able to provide an improvement of 88.26 % and 25.28 % in terms of throughput over SSS and iterative algorithm respectively. The exhaustive optimization provided the best performance i.e., an improvement of around 37.35 % in throughput over iterative algorithm at the cost of high computational complexity. The computational complexity of RL has a quadratic relation with number of users, whereas the computational complexity for exhaustive search increases exponential with number of user which is impractical for real life scenarios. However, in terms of users satisfaction, the RL scheme can achieve a similar performance as exhaustive search with a lower complexity i.e., for 5 users and RWP mobility model, RL is able to provide full user satisfaction to 96 % of the users, which is close to exhaustive search that provides full user satisfaction to 98 % of users. Furthermore, the performance trends of RL compared to other schemes remained unchanged for different mobility models, which proves the robustness of the proposed scheme. In future work, we aim to include the concept of knowledge transfer in our algorithm to further improve its performance and convergence speed.

ACKNOWLEDGMENT

Rizwana Ahmad would like to thank Intel India for Ph.D fellowship. M. D. Soltani and M. Safari acknowledge financial support from the Engineering and Physical Sciences Research Council (EPSRC) under program grant EP/S016570/1 'Terabit Bidirectional Multi-User Optical Wireless System (TOWS) for 6G LiFi'.

REFERENCES

[1] "Cisco visual networking index: Forecast and trends, 2017–2022," White Paper, 2018.

[2] P. H. Pathak, X. Feng, P. Hu, and P. Mohapatra, "Visible light communication, networking, and sensing: A survey, potential and challenges," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 4, pp. 2047–2077, 4th Quart., 2015.

[3] A. Jovicic, J. Li, and T. Richardson, "Visible light communication: Opportunities, challenges and the path to market," *IEEE Commun. Mag.*, vol. 51, no. 12, pp. 26–32, Dec. 2013.

[4] L. Grobe, A. Paraskevopoulos, J. Hilt, D. Schulz, F. Lassak, F. Hartlieb, C. Kottke, V. Jungnickel, and K.-D. Langer, "High-speed visible light communication systems," *IEEE Commun. Mag.*, vol. 51, no. 11, pp. 60–66, Dec. 2013.

[5] H. Haas, "LiFi: Conceptions, misconceptions and opportunities," in *Proc. IEEE Photon. Conf. (IPC)*, Oct. 2016, pp. 680–681.

[6] H. Haas, C. Chen, and D. O'Brien, "A guide to wireless networking by light," *Prog. Quantum Electron.*, vol. 55, pp. 88–111, Sep. 2017.

[7] M. Z. Chowdhury, M. K. Hasan, M. Shahjalal, M. T. Hossan, and Y. M. Jang, "Optical wireless hybrid networks: Trends, opportunities, challenges, and research directions," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 2, pp. 930–966, 2nd Quart., 2020.

[8] M. D. Soltani, X. Wu, M. Safari, and H. Haas, "Bidirectional user throughput maximization based on feedback reduction in LiFi networks," *IEEE Trans. Commun.*, vol. 66, no. 7, pp. 3172–3186, Jul. 2018.

[9] D. Tsonev, H. Chun, S. Rajbhandari, J. J. D. McKendry, S. Videv, E. Gu, M. Haji, S. Watson, A. E. Kelly, G. Faulkner, M. D. Dawson, H. Haas, and D. O'Brien, "A 3-Gb/s single-LED OFDM-based wireless VLC link using a gallium nitride μLED," *IEEE Photon. Technol. Lett.*, vol. 26, no. 7, pp. 637–640, Apr. 1, 2014.

[10] H. Chun, S. Rajbhandari, G. Faulkner, D. Tsonev, E. Xie, E. Gu, J. J. D. McKendry, M. D. Dawson, D. C. O'Brien, and H. Haas, "LED based wavelength division multiplexed 10 Gb/s visible light communications," *J. Lightw. Technol.*, vol. 34, no. 13, pp. 3047–3052, Jul. 1, 2016.

[11] M. Ayyash, H. Elgala, A. Khreishah, V. Jungnickel, T. Little, S. Shao, M. Rahaim, D. Schulz, J. Hilt, and R. Freund, "Coexistence of WiFi and LiFi toward 5G: Concepts, opportunities, and challenges," *IEEE Commun. Mag.*, vol. 54, no. 2, pp. 64–71, Feb. 2016.

[12] M. Z. Chowdhury, M. T. Hossan, A. Islam, and Y. M. Jang, "A comparative survey of optical wireless technologies: Architectures and applications," *IEEE Access*, vol. 6, pp. 9819–9840, 2018.

[13] H. Haas, L. Yin, C. Chen, S. Videv, D. Parol, E. Poves, H. Alshaer, and M. S. Islim, "Introduction to indoor networking concepts and challenges in LiFi," *J. Opt. Commun. Netw.*, vol. 12, no. 2, pp. A190–A203, 2020.

[14] W. Zhang, L. Chen, X. Chen, Z. Yu, Z. Li, and W. Wang, "Design and realization of indoor VLC-Wi-Fi hybrid network," *J. Commun. Inf. Netw.*, vol. 2, no. 4, pp. 75–87, Dec. 2017, doi: 10.1007/s41650-017-0039-1.

[15] M. B. Rahaim, A. M. Vegni, and T. D. C. Little, "A hybrid radio frequency and broadcast visible light communication system," in *Proc. IEEE GLOBECOM Workshops (GC Wkshps)*, Dec. 2011, pp. 792–796.

[16] X. Wu, M. Dehghani Soltani, L. Zhou, M. Safari, and H. Haas, "Hybrid LiFi and WiFi networks: A survey," 2020, *arXiv:2001.04840*. [Online]. Available: http://arxiv.org/abs/2001.04840

[17] W. Ma and L. Zhang, "QoE-driven optimized load balancing design for hybrid LiFi and WiFi networks," *IEEE Commun. Lett.*, vol. 22, no. 11, pp. 2354–2357, Nov. 2018.

[18] Y. Wang and H. Haas, "Dynamic load balancing with handover in hybrid Li-Fi and Wi-Fi networks," *J. Lightw. Technol.*, vol. 33, no. 22, pp. 4671–4682, Nov. 15, 2015.

[19] Y. Wang, D. A. Basnayaka, X. Wu, and H. Haas, "Optimization of load balancing in hybrid LiFi/RF networks," *IEEE Trans. Commun.*, vol. 65, no. 4, pp. 1708–1720, Apr. 2017.

[20] X. Wu, M. Safari, and H. Haas, "Joint optimisation of load balancing and handover for hybrid LiFi and WiFi networks," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Mar. 2017, pp. 1–5.

[21] V. K. Papanikolaou, P. P. Bamidis, P. D. Diamantoulakis, and G. K. Karagiannidis, "Li-Fi and Wi-Fi with common backhaul: Coordination and resource allocation," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2018, pp. 1–6.

[22] M. Kashef, M. Ismail, M. Abdallah, K. A. Qaraqe, and E. Serpedin, "Energy efficient resource allocation for mixed RF/VLC heterogeneous wireless networks," *IEEE J. Sel. Areas Commun.*, vol. 34, no. 4, pp. 883–893, Apr. 2016.

[23] M. S. Demir, S. M. Sait, and M. Uysal, "Unified resource allocation and mobility management technique using particle swarm optimization for VLC networks," *IEEE Photon. J.*, vol. 10, no. 6, pp. 1–9, Dec. 2018.

[24] Y. Wang, X. Wu, and H. Haas, "Fuzzy logic based dynamic handover scheme for indoor Li-Fi and RF hybrid network," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2016, pp. 1–6.

[25] X. Wu, M. Safari, and H. Haas, "Access point selection for hybrid Li-Fi and Wi-Fi networks," *IEEE Trans. Commun.*, vol. 65, no. 12, pp. 5375–5385, Dec. 2017.

[26] R. Zhang, Y. Cui, H. Claussen, H. Haas, and L. Hanzo, "Anticipatory association for indoor visible light communications: Light, follow me!" *IEEE Trans. Wireless Commun.*, vol. 17, no. 4, pp. 2499–2510, Apr. 2018.

[27] C. Wang, G. Wu, and Z. Du, "Reinforcement learning based network selection for hybrid VLC and RF systems," in *Proc. MATEC Web Conf.*, vol. 173. Les Ulis, France: EDP Sciences, 2018, p. 03014.

[28] J. Wang, C. Jiang, H. Zhang, X. Zhang, V. C. M. Leung, and L. Hanzo, "Learning-aided network association for hybrid indoor LiFi-WiFi systems," *IEEE Trans. Veh. Technol.*, vol. 67, no. 4, pp. 3561–3574, Apr. 2018.

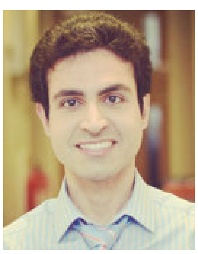
[29] H. Schulze, "Frequency-domain simulation of the indoor wireless optical communication channel," *IEEE Trans. Commun.*, vol. 64, no. 6, pp. 2551–2562, Jun. 2016.

[30] C. Sun, X. Gao, J. Wang, Z. Ding, and X.-G. Xia, "Beam domain massive MIMO for optical wireless communications with transmit lens," *IEEE Trans. Commun.*, vol. 67, no. 3, pp. 2188–2202, Mar. 2019.

[31] S. Ma, R. Yang, H. Li, Z.-L. Dong, H. Gu, and S. Li, "Achievable rate with closed-form for SISO channel and broadcast channel in visible light communication networks," *J. Lightw. Technol.*, vol. 35, no. 14, pp. 2778–2787, Jul. 15, 2017.

[32] C. Bettstetter and C. Wagner, "The spatial node distribution of the random waypoint mobility model," in *Proc. Mobile Ad-Hoc Netzwerke, Deutscher Workshop Über Mobile Ad-Hoc Netzwerke (WMAN)*, vol. 11, 2002, pp. 41–58.
[33] R. Bellman, "A Markovian decision process," *Indiana Univ. Math. J.*, vol. 6, no. 4, pp. 679–684, 1957.
[34] J. Schulman, S. Levine, P. Abbeel, M. Jordan, and P. Moritz, "Trust region policy optimization," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1889–1897.
[35] T. Zhao, H. Hachiya, G. Niu, and M. Sugiyama, "Analysis and improvement of policy gradient estimation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2011, pp. 262–270.
[36] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*. Cambridge, MA, USA: MIT Press, 2018.
[37] T. Kurutach, I. Clavera, Y. Duan, A. Tamar, and P. Abbeel, "Model-ensemble trust-region policy optimization," 2018, *arXiv:1802.10592*. [Online]. Available: http://arxiv.org/abs/1802.10592
[38] Y. Tang and S. Agrawal, "Boosting trust region policy optimization by normalizing flows policy," 2018, *arXiv:1809.10326*. [Online]. Available: http://arxiv.org/abs/1809.10326
[39] O. Nachum, M. Norouzi, K. Xu, and D. Schuurmans, "Trust-PCL: An off-policy trust region method for continuous control," 2017, *arXiv:1707.01891*. [Online]. Available: http://arxiv.org/abs/1707.01891
[40] X. Wu and H. Haas, "Access point assignment in hybrid LiFi and WiFi networks in consideration of LiFi channel blockage," in *Proc. IEEE 18th Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, Jul. 2017, pp. 1–5.
[41] A. Hill, A. Raffin, M. Ernestus, A. Gleave, A. Kanervisto, R. Traore, P. Dhariwal, C. Hesse, O. Klimov, A. Nichol, M. Plappert, A. Radford, J. Schulman, S. Sidor, and Y. Wu. (2018). *Stable Baselines*. [Online]. Available: https://github.com/hill-a/stable-baselines
[42] R. Jain, A. Durresi, and G. Babic, "Throughput fairness index: An explanation," *ATM Forum contribution*, vol. 99, no. 45, pp. 1–13, 1999.

RIZWANA AHMAD (Student Member, IEEE) received the B.Tech. and M.Tech. degrees from the Department of Electronics and Communication Engineering, Aligarh Muslim University, India, in 2014 and 2016, respectively. She is currently pursuing the Ph.D. degree with the Department of Electronics and Communication Engineering, IIIT-Delhi. She is also an Intel Research Fellow at IIIT-Delhi. Her research interests include visible light communication, light fidelity (LiFi), and coexistence of WiFi and LiFi networks.



MOHAMMAD DEHGHANI SOLTANI received the M.Sc. degree from the Department of Electrical Engineering, Amirkabir University of Technology, Tehran, Iran, in 2012, and the Ph.D. degree in electrical engineering from The University of Edinburgh, Edinburgh, U.K., in 2019. During his M.Sc., he was studying wireless communications, MIMO coding, and low complexity design of MIMO-OFDM systems. He worked for two years in the telecommunication industry in Iran. He is currently a Research Associate with the LiFi Research and Development Centre, The University of Edinburgh, funded by the British Engineering and Physical Sciences Research Council. His current research interests include mobility and handover management in wireless cellular networks, optical wireless communications, visible light communications, and LiFi.

MAJID SAFARI (Member, IEEE) received the B.Sc. degree in electrical and computer engineering from the University of Tehran, Iran, in 2003, the M.Sc. degree in electrical engineering from the Sharif University of Technology, Iran, in 2005, and the Ph.D. degree in electrical and computer engineering from the University of Waterloo, Canada, in 2011. He is currently a Senior Lecturer (Associate Professor) with the Institute for Digital Communications, the University of Edinburgh. Before joining Edinburgh in 2013, he held a Postdoctoral Fellowship at McMaster University, Canada. His main research interest is the application of information theory and signal processing in optical communications, including fiber-optic communication, free-space optical communication, visible light communication, and quantum communication. He was the TPC Co-Chair of the 4th International Workshop on Optical Wireless Communication, in 2015. He is currently an Associate Editor of the IEEE Transactions on Communication.

ANAND SRIVASTAVA (Member, IEEE) received the M.Tech. degree and the Ph.D. degree from IIT Delhi, India, in 2002. Before joining IIIT-Delhi in 2014, he was the Dean and Professor of the School of Computing and Electrical Engineering, IIT Mandi, India, from 2012 to 2014, and has been an Adjunct Professor at IIT Delhi, since 2008. Prior to this, he worked at Alcatel-Lucent-Bell Labs, India, from 2009 to 2012, and at the Center for Development of Telematics (CDOT), a telecom research center of the Government of India, from 1989 to 2008. He was also closely involved with ITU-T, Geneva, in Study Group 15, and represented India for various optical networking standards meetings. His current research work is in the area of the optical core and access networks, fiber-wireless (FiWi) architectures, visible light communications (VLC), optical signal processing, and multi-band elastic optical networks (EON).

ABIR DAS received the B.E. degree in electrical engineering from Jadavpur University, India, in 2007, and the M.S. and Ph.D. degrees in electrical engineering from the University of California, Riverside, CA, USA, in 2013 and 2015, respectively. He was a Postdoctoral Researcher at the Computer Science Department, Boston University. He is currently an Assistant Professor with the Computer Science and Engineering Department, IIT Kharagpur, India, and also the Director of the Computer Vision and Intelligence Research (CVIR) Group. His main research interests include computer vision, reinforcement learning, explainable AI, and bias in machine learning.